



Addressing Urban Infrastructure Issues: Solutions for Nairobi and Mexico City

Cao Chengwei 2024300901016
Li Siqi 2024300906007



Background

Nairobi Problem: Matatu Congestion

What are Matatus?

Privately operated minibuses and buses

Main public transport in Nairobi

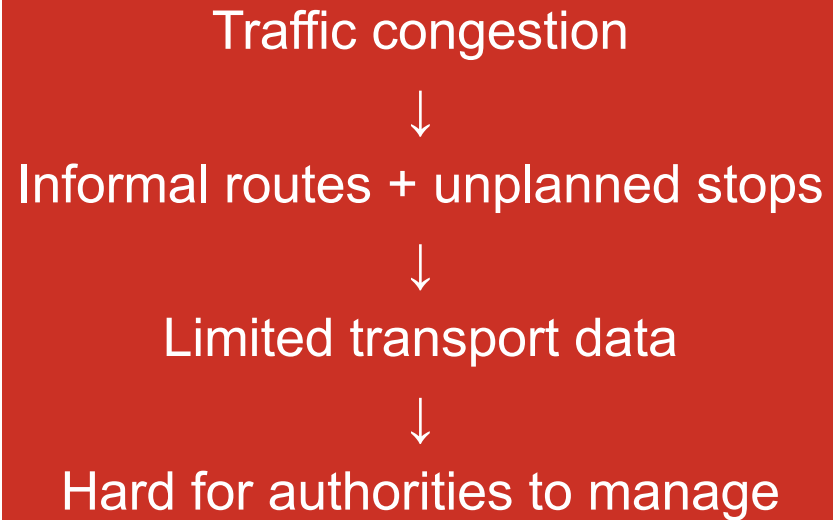


Image 1&2: Nairobi matatu&traffic congestion



Solutions

Two Data-Based Approaches to Reducing Congestion

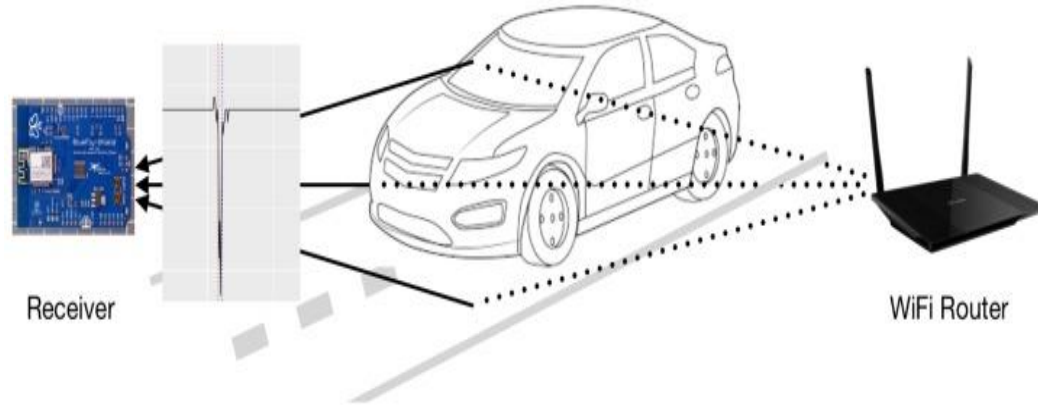


Image 3: Passive RF Sensing & deployment

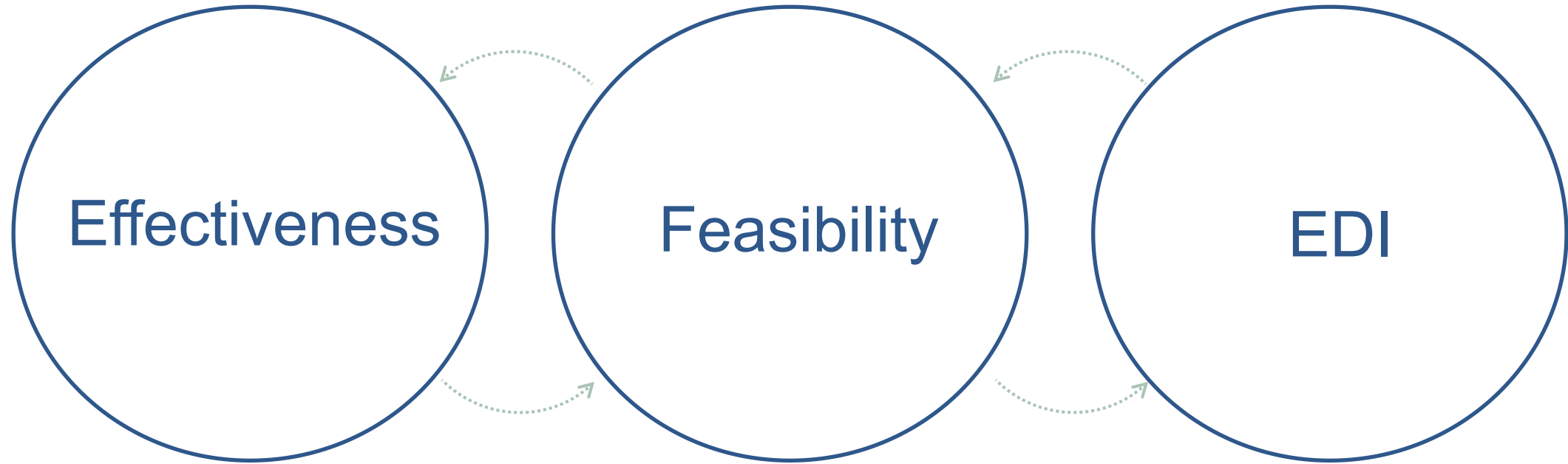


Image 4: App-based GPS Tracking



Evaluation and Recommendation

Two Data-Based Approaches to Reducing Congestion



Land Subsidence in Mexico City

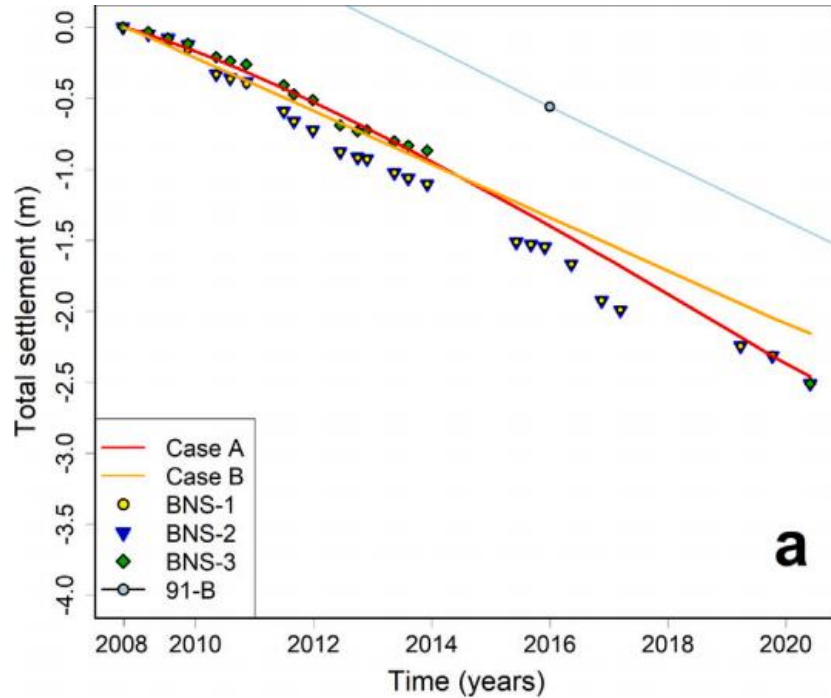


Image 5: Total settlement per year[1]

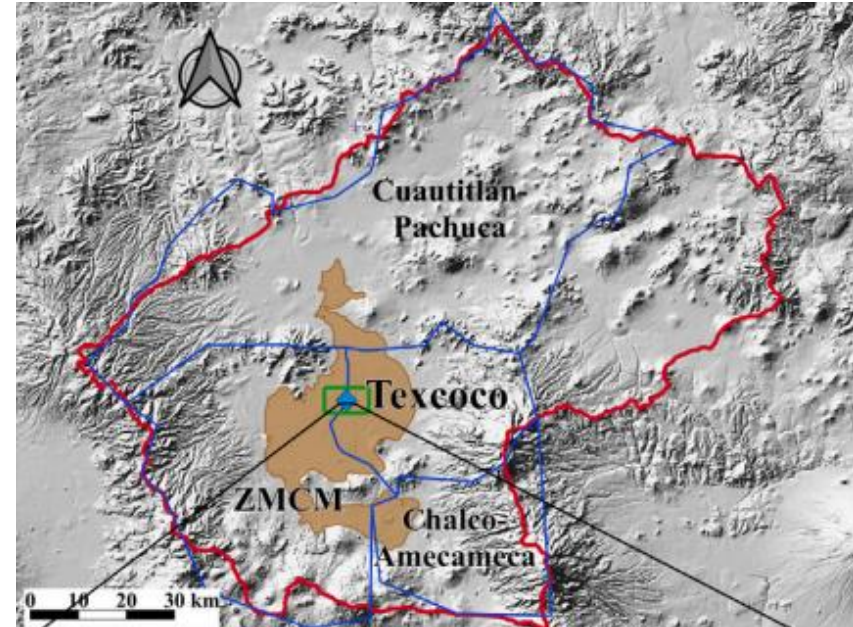


Image 6: Landform of Mexico City[1]

- The average rate of subsidence reaches 0.314m/year[1]
- The main reason for land subsidence is groundwater pumping[1]

Solutions

1. Deep Soil Grouting

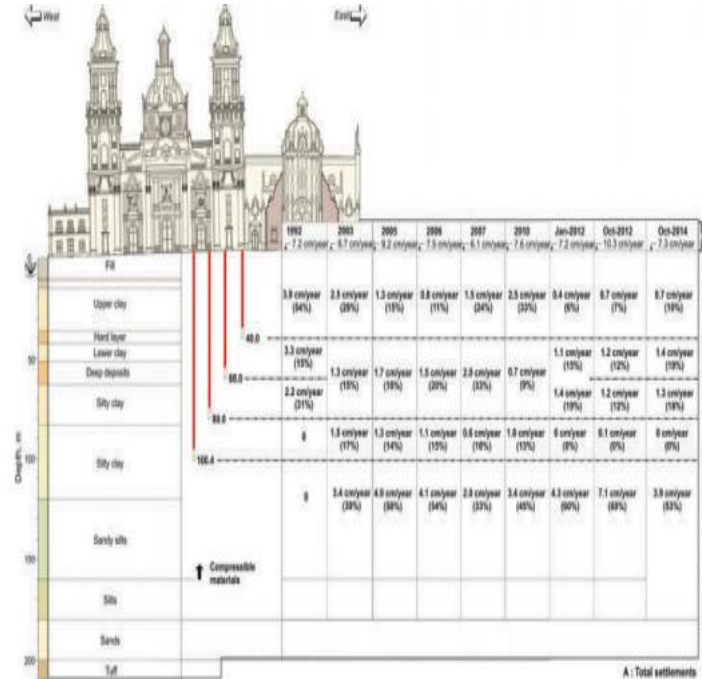
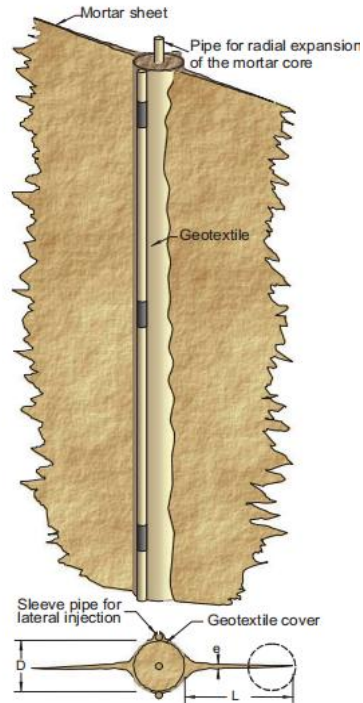


Image 7&8[2]

2. Artificial Aquifer Recharge(AAR)

Wastewater collection and recycling



Artificial recharge of aquifer



Rising groundwater levels



Increasing the support force

Evaluations: Why AAR is better

- Scalable for city-wide deployment
- Lower long-term cost
- Tackle root hydrological cause
- Enhance future water security

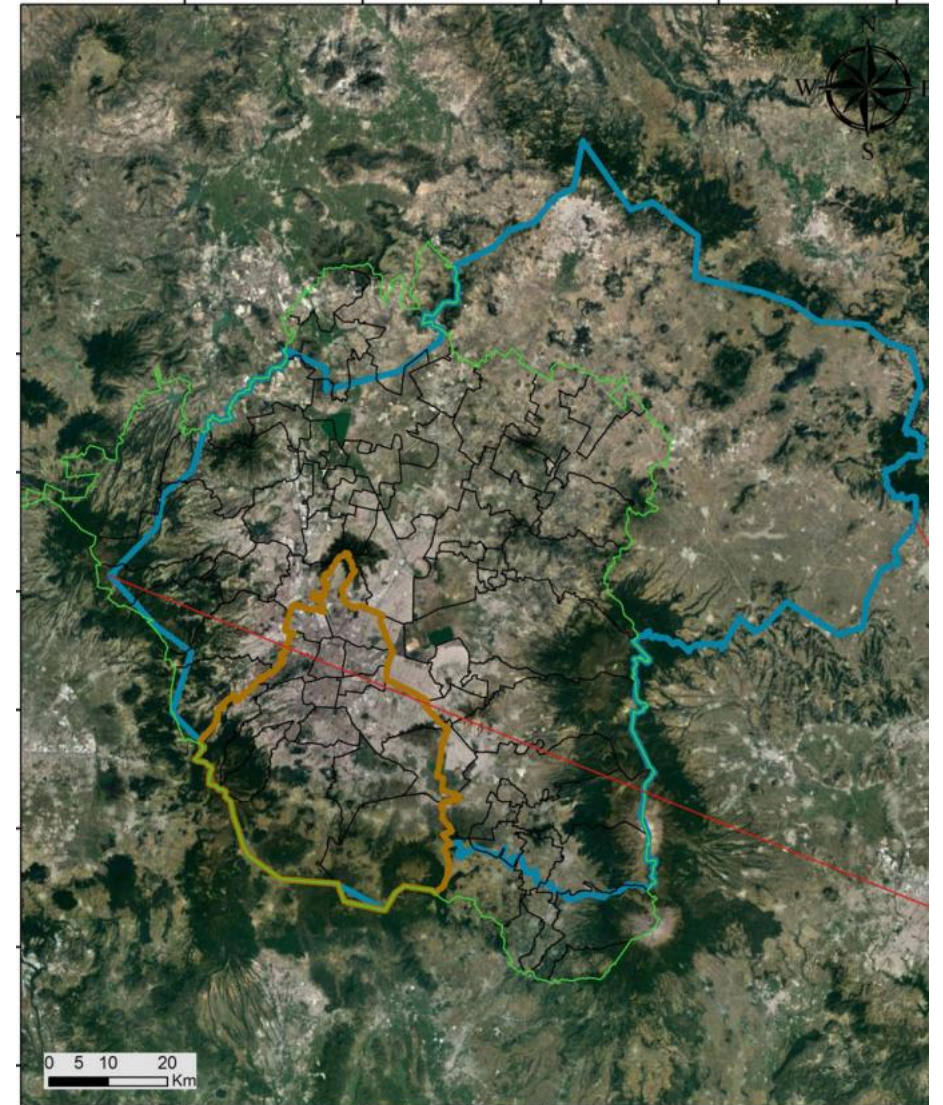


Image 9[3]



Reflection: Different Problems, Similar Thinking

Difference

Congestion vs. subsidence

Similarity

Finding hidden systems

Lesson

Long-term city improvement

Future

Long-term system improvement



Reflection: Engineering for EDI goals



Image 10

- EDI insight: Digital Divide vs Resource Access
- Lesson: Technologies cannot ignore the vulnerable
- Future Goal: Solutions for everyone, not just the resourced



THANK YOU
for watching



Image Sources

Image 10: Microsoft Bing, Accessed: Jun.11,2026, [Online], Available:
https://cn.bing.com/images/search?view=detailV2&ccid=3WF%2F7bvb&id=B0003D56C57D5EC890D18E452AE471539EC375C3&thid=OIP.3WF_7bvbXJ_4w8ZsYqbNLQHaFj&mediaurl=https%3A%2F%2Fpic.ibaotu.com%2F23%2F05%2F18%2Fyay%2F12%2F25656498_b56cd74.jpg%21ww7002&exph=525&expw=700&q=%E7%A4%BE%E4%BC%9A%E5%B9%B3%E7%AD%89%E8%8B%B1%E6%96%87%E5%9B%BE%E7%89%87&FORM=IRPRST&ck=0C020A5B423C76AACCE4EE33F7526130&selectedIndex=31&itb=0&cw=1529&ch=842&ajaxhist=0&ajaxserp=0



References

- [1] B. Zapata-Norberto, E. Morales-Casique, R. Contreras-Galvan and J.A. Ramos-Leal, “Land subsidence in Mexico City: New insights from field data and numerical modeling,” *Groundwater for Sustainable Development*, vol.29, p. 101432, Mar 2025, doi: 10.1016/j.gsd.2025.101432
- [2] E. Ovando- Shelley, E. Santoyo- Villa, and J. Hernández, “Mexico City’s Metropolitan Cathedral and Sagrario Church 13 Years After Underexcavation and Soil Hardening,” *International Journal of Architectural Heritage*, vol. 10, nos. 2–3, pp. 346–359, 2016, doi: 10.1080/15583058.2015.1113331.
- [3] A. P. Nava, T. K. Parker, and R. B. C. Paredes, “Challenges and experiences of managed aquifer recharge in the Mexico City Metropolitan Area,” *Groundwater*, vol. 60, no. 5, pp. 675–684, 2022, doi:10.1111/gwat.13237.